2010 F=ma Exam: Problem 25

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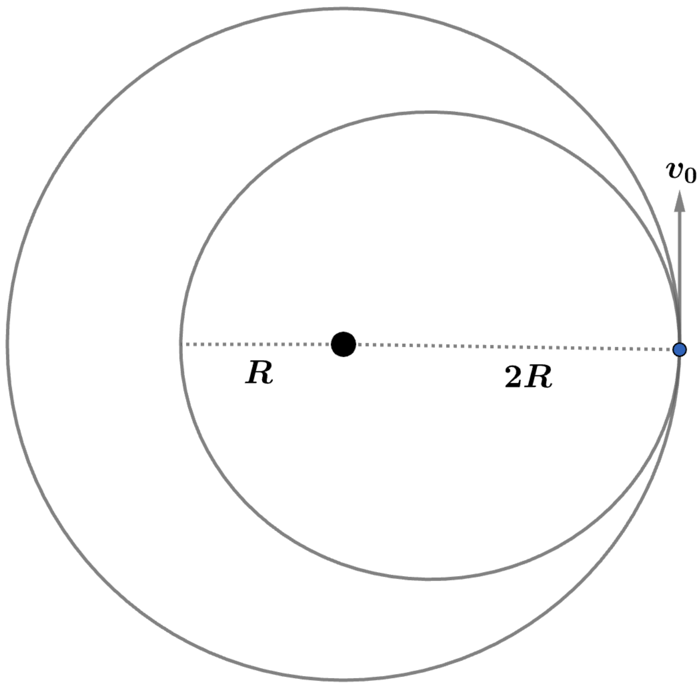

Recall from the elliptical orbit equations that

$$v_{\text{apogee}} = \frac{a-c}{b}\sqrt{\frac{GM}{a}}$$

In our case, we have

$$R = a - c$$
$$2R = a + c$$

Adding the equations,

$$3R = 2a$$
$$a = \frac{3R}{2}$$

Multiplying the equations,

$$2R^2 = a^2 - c^2 = b^2$$
$$b = \sqrt{2}R$$

Substituting into the expression for v_{apogee},

$$v_0 = \frac{R}{\sqrt{2}R}\sqrt{\frac{GM}{3R/2}} = \sqrt{\frac{GM}{3R}}$$

The velocity of a circular orbit of radius $2R$ is

$$v_1 = \sqrt{\frac{GM}{2R}} = \sqrt{\frac{3}{2}}v_0$$

so the answer is $\boxed{\text{A}}$.